

DAY 18

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Task:

- Create a small web api project that fetches list of pets (dog varieties) from SQL server via code-first Entity framework approach

Note: THIS SHOULD BE completely done in VS CODE.

Code:

Dog.cs (Model)

```
using System.ComponentModel.DataAnnotations;
```

```
namespace Day19Task.Models
{
    public class Dog
    {
        public int Id { get; set; }

        [Required]
        public string? Breed { get; set; }

        public string? Origin { get; set; }
    }
}
```

AppDbContext.cs (Context)

```
using Microsoft.EntityFrameworkCore;
using Day19Task.Models;
```

```
namespace Day19Task.Context
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }
    }
}
```

```

        public DbSet<Dog> Dogs { get; set; }
    }
}

```

appsettings.json

```

{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "ConnectionStrings": {
    "DefaultConnection":
"Server=PTSQLTESTDB01;Database=Nandhini_Demo;Trusted_Connection=True;TrustServerCertificate=True;"
  },
  "AllowedHosts": "*"
}

```

Program.cs

```

using Day19Task.Context;
using Microsoft.EntityFrameworkCore;
using Microsoft.OpenApi.Models;
var builder = WebApplication.CreateBuilder(args);

builder.Services.AddOpenApi();
builder.Services.AddControllers();
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnection")));

var app = builder.Build();

if (app.Environment.IsDevelopment())
{
    app.UseSwagger();
}

```

```

    app.UseSwaggerUI();
}

if (app.Environment.IsDevelopment())
{
    app.MapOpenApi();
}

app.UseHttpsRedirection();
app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();

app.Run();

```

DogController.cs (Controller)

```

using Microsoft.AspNetCore.Mvc;
using Day19Task.Context;
using Day19Task.Models;

namespace MyAspNetApp.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class DogController : ControllerBase
    {
        private readonly AppDbContext _context;

        public DogController(AppDbContext context)
        {
            _context = context;
        }

        [HttpGet("FetchAllDogs")]
        public IActionResult GetDogs()
        {
            var dogs = _context.Dogs.ToList();
            return Ok(dogs);
        }

        [HttpPost("AddDog")]
        public IActionResult AddDog(Dog dog)

```

```

    {
        _context.Dogs.Add(dog);
        _context.SaveChanges();
        return Ok("Dog added");
    }
}

```

Output:

Before adding - Database

Results

Messages

	Id	Breed	Origin
1	1	Labrador	Canada
2	2	German Shepherd	Germany
3	3	Pug	China

Adding values and its response

Request body

Edit Value | Schema

```

{
  "id": 0,
  "breed": "Golden Retriever",
  "origin": "Scotland"
}

```

Request URL

<http://localhost:5281/api/Dog/AddDog>

Server response

Code

Details

200

Response body

Dog added

After adding - Database

Results		Messages	
	Id	Breed	Origin
1	1	Labrador	Canada
2	2	German Shepherd	Germany
3	3	Pug	China
4	4	Golden Retriever	Scotland

Fetching and displaying all dogs from the database

Request URL

http://localhost:5281/api/Dog/FetchAllDogs

Server response

Code

Details

200

Response body

```
[
  {
    "id": 1,
    "breed": "Labrador",
    "origin": "Canada"
  },
  {
    "id": 2,
    "breed": "German Shepherd",
    "origin": "Germany"
  },
  {
    "id": 3,
    "breed": "Pug",
    "origin": "China"
  },
  {
    "id": 4,
    "breed": "Golden Retriever",
    "origin": "Scotland"
  }
]
```